RewardBook - Code Changes

Debugging summary | October 30, 2026

Overview

Fixed reward calculation boundaries, duplicate event processing, refund validation, and incomplete repeat-alert records. Added regression tests and recorded the test evidence.

What changed

Points calculation: validates non-negative integer amounts and reward rules; a cap of zero is now honored. Existing integer floor-rounding behavior in the working copy was preserved, not introduced by this debugging pass.

Event idempotency: missing or invalid event IDs are rejected, and already-processed IDs are ignored so a replay cannot change points or add another ledger entry.

Purchase validation: captured purchases require a valid transaction ID and a non-negative integer amount.

Refund safety: refunds must be positive integer amounts and fit within the purchase amount still available to refund. Reversals use integer proportional arithmetic, cannot exceed awarded points, and cannot reduce the balance below zero.

Repeat alerts: rows with missing/invalid transaction IDs, account IDs, dates, or amounts are skipped instead of being grouped into alerts.

Regression tests added

- **test_refund_event_replay_is_idempotent** - a repeated refund is applied only once.
- **test_refund_cannot_exceed_remaining_purchase_amount** - a second refund cannot exceed the remaining amount.
- **test_repeat_alert_skips_rows_missing_identifiers** - incomplete rows do not create an alert.

Verification

- Baseline: **47 failed, 167 passed**.
- Final: **python -m pytest -q** - **217 passed** on Python 3.14.6.
- **python -m main** ran successfully: Points example: 27; Repeat warnings: [].
- Pylance found no diagnostics for service.py; syntax checks passed for changed Python files; **git diff --check** reported no whitespace errors.

A short explanation you can use

> ?I fixed zero-cap handling, made purchase and refund events safe to replay, validated refunds so they cannot exceed the purchase or create a negative points balance, and prevented repeat alerts from using incomplete records. I added regression tests, and all 217 tests pass.?

Files changed

File	Summary
service.py	Input validation, zero-cap support, event idempotency, refund limits/reversal safety, and repeat-alert filtering.
tests/test_acceptance_matrix.py	Three regression tests for replayed refunds, cumulative refund limits, and missing alert identifiers.
TESTING.md	Environment/test results, edge-case evidence, and regression test notes.

This PDF summarizes the debugging edits in this session. Unrelated pre-existing untracked build metadata was not included.